Professional Elective II

List of Courses

Course Code	Course Title	Sem	Category	Periods/Week				C
				L	T	P	S	
CSC1371	**APPLICATION DEVELOPMENT**	**VI**	**PE**	**2**	**0**	**2**	**2**	**3**
Pre-requisite	Web Application Development, Java Programming	**Course Type**	**Theory (PRBL)**	**QP TYPE**				**QP1**

Course Objectives:

To Develop the skills to create native applications with basic GUI components for intuitive user interactions, build cross-platform applications with effective event-handling mechanisms, and integrate location services and data storage to enhance functionality. Additionally, learn to develop web applications with seamless database access for efficient data management.

Course Outcomes:

Upon completion of the course, the students will be able to:

CO	Course Outcomes
CO1	Gain proficiency in developing native applications with graphical user interface (GUI) components.
CO2	Learn to create hybrid applications incorporating basic event-handling mechanisms.
CO3	Implement cross-platform applications with integrated location services and data storage capabilities.
CO4	Analyze the design and development of cross-platform applications by examining GUI elements and event-handling functionality.
CO5	Understand and apply fundamental principles to enhance app performance, maintainability, UI/UX design, and basic security measures.

SYLLABUS

Module I	**BASICS OF MOBILE AND WEB APPLICATION DEVELOPMENT**	**6+6**

Introduction to Web and Mobile Application Development: Native, Hybrid, and Cross-Platform Apps.
Progressive Web Apps, Principles of Responsive Web Design.

Guided Self-Study Topics		
Application Programming Interfaces (APIs) in Mobile and Web Development, Mobile Backend as a Service (MBaaS) and Its Role in Modern App Development.		
Module II	**NATIVE APP DEVELOPMENT USING JAVA**	**6+6**
Native Web App, Scenarios to create Native App, Tools for creating Native App, Popular Native App Development Frameworks. Java & Kotlin for Android, Swift & Objective –C for iOS, Basics of React Native, Native Components, JSX, State, Props.		
Guided Self-Study Topics		
Flutter for Cross-Platform App Development-App Performance Optimization Techniques.		
Module III	**HYBRID APP DEVELOPMENT**	**6+6**
Hybrid Web Apps: Development Criteria, Tools, Challenges. Popular Frameworks like Ionic and Apache Cordova.		
Guided Self-Study Topics		
Comparison of Hybrid, Native, and Cross-Platform App Development- Progressive Web Apps (PWAs) and Their Role in Modern Web Development.		
Module IV	**CROSS-PLATFORM APPLICATION DEVELOPMENT USING REACT NATIVE**	**6+6**
Cross-Platform App Development, React Native Architecture, Layouts, Styling, and Navigation, Event Handling and Data Flow. Introduction to Flutter, Architecture, Widgets, Layouts and Styling, Event Handling and User Interaction.		
Guided Self-Study Topics		
State Management in Cross-Platform Applications.		
Module V	**ESSENTIAL APP PERFORMANCE AND MAINTENANCE**	**6+6**
App Performance, Debugging, Maintainability, UI/UX Essentials. App Security, Performance Optimization Techniques.		
Guided Self-Study Topics		
Security Considerations in Mobile and Web App Development.		
		Total Hours: 60 Hours
Learning Resources:		
Text Books:		
1	Head First Android Development, Dawn Griffiths, O'Reilly ,3rd edition, 2021.	

2	Apache Cordova in Action" by Raymond K. Camden, Manning, 2019.
Reference Books:	
1	Android Programming for Beginners" by John Horton, Packt Publishing, 2nd Edition.
2	Native Mobile Development" by Shaun Lewis and Mike Dunn.
3	Building Cross-Platform Mobile and Web Apps for Engineers and Scientists: An Active Learning Approach" by Pawan Lingras, Matt Triff, Rucha Lingras.
4	Apache Cordova 4 Programming" by John M. Wargo, 2015.
5	React Native Cookbook" by Daniel Ward, Packt Publishing, 2nd Edition.